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Enchanting performance measurement and management with data envelopment analysis: Insights from bibliometric data visualization and analysis

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ABSTRACT

While the breadth of applicative uses of data envelopment analysis (DEA) has increased over time, this non-parametric frontier methodology has proven effective for measuring how efficient decision-making units (DMUs) are. One of its uses is performance measurement in various organizations — from universities and banks to investment funds and entrepreneurial small and medium-sized enterprises. Following a search inquiry in the Scopus and Web of Science databases, we found no bibliometric analysis and keyword co-occurrence network visualization of just the use of DEA for performance management. With this study, we attempt to investigate the DEA literature for performance measurement and management, considering articles in scientific peer-reviewed journals, which are indexed in the Web of Science database. The analysis and visualization of the bibliometric data are based on 89 articles from the start of the century onward. We present the annual trends of published articles, journals, authors, and countries leading the scientific discussion and citation analysis. A detailed keyword visualization analysis was performed and is discussed in this article alongside the paths for future hotspots of the use of DEA for performance management. The findings have implications for both the academic community and practitioners, business leaders, managers, and policy-makers.

1. Introduction

Data envelopment analysis (DEA) publications have increased significantly since the seminal paper by Charnes et al. [1]. This was the first time this methodology was established as a nonparametric frontier approach for analyzing how efficient decision-making units (DMUs) are. Using Farell's concept of efficiency, Charnes et al. [1] established the mathematical modeling methodology DEA with an eye toward analyzing real-world instances of the growth of efficiency in elementary schools.

Thus, it is concluded that it is possible to identify and evaluate the areas of optimal performance or best practice within the sample using the non-parametric frontier approach known as DEA. For all organizations, efficiency evaluation is essential. While managers can assess the profitability of an organization by utilizing financial indicators, efficiency assessment becomes more challenging when managers start employing numerous comparable inputs and multiple commensurate outputs [2].

With data envelopment analysis, the major goal is to assess each decision-making unit's efficiency in comparison to other units in the observational set [3,4]. All DMUs are assumed to use the same inputs and generate the same outputs [5]. The areas of its applicability have been greatly expanded since 1978. Illustratively, in the most recent four decades from the end of the 1970s until the middle of 2010s, more than 10,300 DEA-related articles have been published in the literature [6]. According to Emrouznejad and Yang [6], supply chains, public policy-making, agriculture, finances, and transportation are the areas where DEA is most frequently used. With that in mind, assessing the efficiency of the performance measurement systems in an organization plays a crucial role in managerial decision-making. A performance measurement system represents a succinct and defined collection of both financial and non-financial measures that supports an organization's decision-making process by gathering, processing, and evaluating quantified data of performance information [6]. In other words, determining whether the organizational plan is achieved is the most crucial role of performance measurement.

As a topic, the interest in performance measurement is growing. To illustrate, it is noted that nearly 4000 publications on performance measurement were published between 1994 and 1996, giving rise to the term 'performance measurement revolution' [7]. Performance management and measuring procedures are now standard across all spheres of business and industry, including the non-profit and public sectors [2]. We are also witnessing swift changes in how we operate

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organizations, which are being fueled by quickly evolving technology, growing globalization, and the removal of trade barriers. High and sustained earnings from a well-run business can lead to increased job prospects and personal income; additionally, a company's financial success will increase employee salaries, result in better manufacturing facilities, and deliver higher-quality goods to clients. Without a result measurement, this procedure is impossible. As a result, performance measurement is crucial for the efficient administration of the company. Moreover, measuring performance is necessary to analyze the management plan, track its status, forecast upcoming situations and trends, and follow the behavior of the members of the organization in relation to the objectives [4]. Ultimately, this serves as a foundation for making choices when they are needed.

We believe a study that analyzes the application of DEA for the purposes of performance measurement and management is needed as it was found that this non-parametric approach is one of the most often used methods for measuring the performance of organizations, organizational units, and segments, as well as employees [8]. Additionally, a gap in the literature was identified after we conducted a query in the Web of Science database of indexed published articles and did not find an article exploring this topic and this particular use of the DEA methodology. This represented an additional motivation for carrying out this research endeavor. Considering this, our article addresses several research questions (RQ):

RQ1: What are the key trends in the application of data envelopment analysis in performance measurement and management?

RQ2: From the start of the century, which peer-reviewed scientific articles, authors, and journals have impacted the scientific discourse the most?

RQ3: What are the connected key and emerging topics and hotspots concerning the specific application of data envelopment analysis?

RQ4: What issues, challenges, and emerging themes dominate the published literature and appear most frequently, and which ones will persist in the future?

To answer these research questions, we acquire the required data by extracting the textual data and keywords from peer-reviewed published articles in academic journals, which are indexed in the Web of Science database as one of the world's leading databases for high-quality global research. After the extraction of the data, we employ the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol for identifying, screening, assessing the eligibility [9], and finally including only the most relevant articles in the subsequent descriptive analytics and bibliometric analysis. We use descriptive analytics for analyzing the trends of published articles detailing the use of DEA for performance measurement, identifying the most prolific and most cited journals, the most productive authors and co-authorship teams, as well as the countries that drive the academic discourse in this topic forward. Moreover, bibliometric techniques and visualization of bibliometric data help us in developing an in-depth content analysis of the most occurring terms in the titles, abstracts, and keywords of scientific articles, detecting research hotspots and recent trends, and pointing out areas for future development of DEA's application in performance measurement activities.

One of the most alluring yet frequently overlooked topics regarding DEA applications is the measurement of organizational performance. Based on a thorough study of the keyword co-occurrence visualization network, we look at the state of published literature detailing the application of DEA in performance measurement activities as well as the development trajectory of this scientific niche. Furthermore, this article is organized in a way that it presents the introduction in the first section, followed by the used methods for the bibliometric analysis in the second section. Then, the article moves on to describing the main findings and results and discussing them in correlation with articles that have used the DEA methodology for assessing performance in the third and fourth sections respectively. The article concludes with directions for future research in the fifth section and the main conclusions in the sixth section.

2. Methodology

A bibliometric review or bibliometric analysis is used as a research approach to examine the evolutionary development of scientific literature through time and shed light on the intellectual relationships and variety of knowledge [10]. Additionally, a descriptive component of the literature study is used to gather new data on the authors and papers that receive the most citations in the area, keyword relationships and occurrences, and comparable topics. The data for the bibliometric analysis has been downloaded from the Web of Science database. We searched for the keywords "performance measurement" AND "data envelopment analysis" OR "DEA". Based on the keywords, we found that there is no review of the literature regarding the niche use of data envelopment analysis that we wanted to analyze, that is, for performance measurement.

During the search query, the year 2023 was excluded as a publishing year because, at the moment of writing, this year is not finished. Additionally, the intention was to analyze only articles in scientific peer-reviewed journals, which are indexed in the Web of Science database. The language of the articles is English. Furthermore, the search was further refined and limited to category management; the citation topics meso was also chosen to be only management. Considering this, 140 articles were identified in the database. All pieces of data related to each article, such as authors, source title (journal), year of publication, title, number of citations from all databases, authors' affiliations, keywords, and abstract, were retrieved from the Web of Science database on May 3, 2023. While a bibliometric analysis is often conducted based on larger sample sizes, smaller datasets can be beneficial for reviewing the published literature, especially in cases of niche and emerging topics like the application of data envelopment analyses in measuring performance [11].

For identifying the relevant data, cleaning it, and including articles for analysis, we used the PRISMA protocol. First, we removed the articles not written in English (five of them), and since there were no duplicates, we undertook a manual check of the relevance of each article based on an analysis of the abstract and the title. Forty-six articles were removed due to irrelevance as these focused on marketing performance, selection of Six Sigma projects, research-and-development-based innovation, opportunity recognition, project management, and similar. A breakdown of the completed steps of identification, screening, eligibility, and inclusion per the PRISMA protocol is presented in Fig. 1.

Furthermore, the VOSviewer program (version 1.6.19) was employed for the keyword co-occurrence analysis. Van Eck and Waltman [12] created this program for building and visualizing networks grounded in bibliometric and textual data. Many different study domains, particularly those pertaining to organizational and managerial sciences, have demonstrated their broad range of applicability. We were able to create network and density visualization maps using the program, as well as graphically display keyword relationships and clusters.

3. Results

3.1. Descriptive overviewing analysis

In this section, we provide an overview of the findings from the bibliometric analysis of articles describing the use of the data envelopment analysis methodology for performance measurement in terms of annual trends, authors with the highest number of published articles and citations, country authorship, scientific journals, and citations. Even though DEA has been used for more than 40 years, the first articles of its use for the purposes of measuring performance in organizations occurred later, in the first decade of the century.

The popularity of this method in performance management is proven by the fact that in 2022, there were 18 articles published

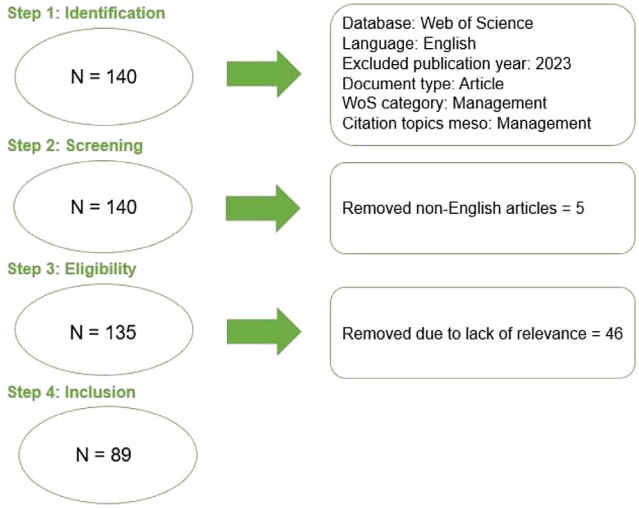

Fig. 1. Visual representation of the applied PRISMA protocol.

Fig. 2. Annual trends of DEA articles for performance management.

Fig. 3. Five-year trends of published DEA articles for performance management.

in peer-reviewed journals, which represents the highest number of published articles in a year so far. In other words, solely in 2022, the number of published articles of DEA for performance measurement exceeded the number of published articles in 2021 and 2020 combined. Moreover, it is evident that the publishing trends have been cyclical, with some years being less focused on this topic than others. Fig. 2 presents the yearly trends.

Fig. 3 provides a closer look at the most recent publishing trends from the last five years. From four articles in peer-reviewed journals in 2018, the number progressed to 18 in 2022, potentially accelerated by the increase in the interest and utilization of the DEA methodology in practice and contemporary decision-making.

Fig. 4. Most-cited individual authors.

We identified a total of 56 journals that have published articles about the use of DEA for performance measurement. These journals were included following a set of criteria, including the highest number of citations attributed to the articles on this topic, the highest number of total published articles, the highest impact factor (2021 release), and the highest SCImago Journal Rankings (2022 release). While some of the data was extracted from the Web of Science database, the last ones were retrieved from the publicly available SCImago Journal & Country Rank online web portal [13].

Based on the extracted dataset, the analysis pointed out the *Journal of Intellectual Capital* as the journal with the most citations and the highest number of published articles, whereas the *Supply Chain Management: An International Journal* currently has the highest impact factor from the analyzed sample (see Table 1).

Out of the 89 articles included in the analysis, 91% of them (81 articles) were cited at least once, while 47.2% (42 articles) had more than ten or double-digit citations. The most-cited articles (Table 2) have all been published in the previously identified journals. The article "Assessing the relative performance of UK university technology transfer offices: parametric and non-parametric evidence" by Chapple et al. [14], published in *Research Policy*, is the most-cited article (297 citations) that researches the analyzed topic. Further, it is one of the first published articles in the field.

By analyzing the authors' research landscape, we identified only six articles, which were written by a single author. So, most of the articles are written by a co-authorship team. Driven by the will to showcase the top authors in the field, who are carrying the research discussion forward, we present the most-cited authors individually (Fig. 4), whereas Fig. 5 provides an overview of the co-authorship teams with the highest number of citations of their published work. The case with most of the co-authorship teams is that the authors who are a part of the teams have collaborated on just one article, so it is not common to find the same co-authorship team behind two or more published articles. The majority of articles, or 34 of them (38.2%), were written by three authors, with the average number of authors being 3.02. Furthermore, the highest number of authors per article is five.

During the exploration of the use of DEA for performance measurement, which has lasted for more than two decades so far, several contexts and even geographical settings have been analyzed. This makes the country authorship landscape exceptionally diverse while still leaving space for further analysis in other regions where this methodology, application, and impact were not previously researched at all or just a little.

To gauge the impact of each country when it comes to published articles, we analyzed the number of published articles and the number of citations. The analysis has shown that the United States dominates the research landscape currently with the highest number of citations and the total number of articles. Evidently, not all countries that are leading in terms of published articles, retain the top spots for the total number of citations, and vice versa.